New classes are noted in blue for illustrative purposes

src.pacman

portals

<<enumeration>> PortalColour
- White
- Yellow
- DarkGold
- DarkGray

+ getImageName() : String

Portal
- location : Location

+ getPair() : Portal
+ setPair(pair: Portal) : void
+ getLocation() : Location
+ getColour() : PortalColour

PortalFacade
- portalPairs : HashMap<PortalColour, ArrayList<Portal>>

+ putPortals(): void
+ setUpPortals(): void
+ putWhitePortal(bg: GGBackground, location: Location) : void
+ putYellowPortal(bg: GGBackground, location: Location) : void
+ putDarkGoldPortal(bg: GGBackground, location: Location) : void
+ putDarkGreyPortal(bg: GGBackground, location: Location) : void
+ findPortalPairs(): void
+ findColourPair(portal: Portal) : void
+ removePortals() : void

levelchecking

AccessibilityCheckerStrategy
- errorMessages : ArrayList<String>
- visited: boolean[][]

+ checkLevel(grid: PacManGameGrid) : void
- floodFill(grid: PacManGameGrid, startX: int, startY: int) : void
- isValidLocation(grid: PacManGameGrid, x: int, y: int) : boolean
- findPortalLocations(grid: PacManGameGrid) : List<Location>

GoldAndPillCountCheckerStrategy
- numGoldAndPills : int
- errorMessage : String = ""

+ checkLevel(grid: PacManGameGrid) : void
- countGoldAndPills(grid: PacManGameGrid) : int

PacManCheckStrategy
- numPacMan : int
- pacActorLocations : ArrayList<Location>
- errorMessage : String = ""

+ checkLevel(grid: PacManGameGrid) : void
- findAllPacmanLocations(grid: PacManGameGrid): ArrayList<Location>

PortalCheckStrategy
- numPacMan : int
- portalMap : HashMap<String,ArrayList<Location>>
- errorMessages : ArrayList<String>

+ checkLevel(grid: PacManGameGrid) : void
- checkPortalCounts(grid: PacManGameGrid): void
- setUpPortalMap(grid: PacManGameGrid): void

LevelChecker
- checkingStrategies : ArrayList<LevelCheckStrategy>
- errorMessages : ArrayList<String>
- isError : boolean = false
- accessibilityChecker : LevelCheckStrategy
- fileName : String

+ doLevelCheck() : boolean
+ getErrorMessage () : ArrayList<String>

<<interface>> LevelCheckStrategy

checkLevel(grid: PacmManGameGrid) : String

ItemHandler
- iceCubes : ArrayList<Actor>
- goldPieces : ArrayList<Actor>
- pillAndItemLocations : ArrayList<Location>
- propertyPillLocations : ArrayList<Location>
- propertyGoldLocations : ArrayList<Location>

+ countPillsAndItems() : int
+ getPillAndItemLocations() : ArrayList<Location>
+ setUpPillAndItemsLocations() : void
+ putPill(bg: GGBackground, location: Location) : void
+ putGold(bg: GGBackground, location: Location) : void
+ pitIce(bg: GGBackground), location: Location) : void
+ removeItem(type: String, location: Location) : void
+ getGoldPieces() : ArrayList<Actor>
+ getPropertyPillLocations(): ArrayList<Location>
+ getPropertyGoldLocations(): ArrayList<Locations>

Game
- bg : GGBackground
- seed : int = 30006
- hasPacManBeenHit : boolean
- hasPacManEatenAllPills : boolean
- properties: Properties
- levelsFolder : String
- currentLevelPath : String
- currentLevel : int
- maxLevel : int
- levelFiles : HashMap<Integer, String>
- passedGameChecking : boolean = true
- passedLevelChecking : boolean = true

- play(currentLevel: int) : void
- playLevel() : void
- gameCheckingAndSetup() : void
- doLevelChecking() : void
- duplicateLevelsError() : void
- setFirstLevel() : void
- getLevelFromFile(fileName: String) : int
- gameSetUp() : void
- lookForCollisions() : void
- checkGameEnd() : void
- setPacActorStart() : void
- setMonsterLocations() : void
- isXmlFile(filePath: String) : boolean
- drawGrid(bg: GGBackground) : void
- hasPacmanBeenHit() : boolean
- hasPacmanEatAllPills() : boolean
- setGameSlowDown() : void
- setCharacterSeeds() : void
+ getNumHorzCells() :int
+ getNumVertCells() : int
+ getItemHandler() : ItemHandler
+ getPacActor() : PacActor
+ getGrid(): PacManGameGrid
+ getPortalFacade() : PortalFacade
+ getGameCallback() : GameCallback

PacManGameGrid
- nbHorzCells : int
- nbVertCells : int
- mazeArray: int[][]
- fileName: String
- filePath: String
- pacActorLocation: Location
- numPacActors : int

- loadMap() : void
+ getCell(location: Location) : int
+ getNumPacActors() : int
+ getFilePath() : String
+ getFileName() : String

Driver
+ DEFAULT_PROPERTIES_PATH: String

+ main(args: String[]) : void

utility

GameCallback
- logFilePath : String = "pacman/Log.txt"
- errorLogFilePath : String = "pacman/ErrorLog.txt"
- fileWriter : FileWriter
- errorFileWriter : FileWriter

+ logErrorMessages(errorMessages : ArrayList<String>) : void
+ writeError(str: String) : void
+ writeString(str: String) : void
+ endOfGame(gameResult: String) : void
+ pacmanLocationChanged(pacmanLocation: Location, score: int, nbPills: int) : void
+ monsterLocationChanged(monster: Monster) : void
+ pacManEatPillsAndItems(pacmanLocation: Location, type: String) : void

PropertiesLoader

+ loadPropertiesFile(propertiesFile: String) : void

ch.aplu.jgamegrid

GameGrid
...

+ toPoint(location: Location) : Point
+ getBg() : GGBackround
+ doRun() : void
+ show() : void
# addActor(actor: Actor, location: Location, direction: double, derefresh: boolean) : void
+ delay(time: long) void

Location
...

+ getX() : int
+ getY() : int
+ get4CompassDirectionTo(location: Location) : CompassDirection
+ setDirection(direction: Double) : void
...

Actor
- location : Location
- direction : double
- idVisible : int

+ act() : void
+ getDirection() : Double
+ setDirection(direction: Double) : void
+ setLocation(location: Location) : void
+ turn(angle: Double) : void
...

PacActor
- idSprite : int
- nbPills : int
- score : int
- visitedList : ArrayList<Location>
- propertyMoves : List<String>
- propertyMoveIndex : int
- <<final>> listLength : int
- randomiser : Random
- isAuto : boolean = false
- destination : Location
- previousLocation : Location
+ hasTeleported : boolean
- useFindClosestStrategy : boolean

+ act() : void
- closestPillLocation() : Location
- eatPill(location: Location) : void
- addVisitedList(location: Location) : void
- isVisited(location: Location) : boolean
- canMove(location: Location) : boolean
- moveInAutoMode() : void
+ keyRepeated(keyCode : int) : void
- followPropertyMoves() : void
+ steppedOnPortal() : Portal
+ hasJustComeFromPair(portal: Portal) : boolean
+ teleport(portal: Portal) : void
+ getPreviousLocation() : Location
+ findNext() : Location
+ findNextWithoutPortal() : Location
+ findClosest(possibleLocations: ArrayList<Location>, destination: Location)
+ needPortal(tileTypes: ArrayList<Integer>) : ArrayList<Location>
+ floorFill(visited: boolean[][]): void
+ isValidLocation(x: int, y: int) : boolean
+ alreadyEaten(location Location) : boolean

monster

MonsterFacade
- monsters : ArrayList<Monster>
- actorMap : Map<String,Actor>
- monsterLocations : Map<Monster, Location>

+ setUpMonsters(game: Game) : void
+ delayTx5(seconds: int) : void
+ setMonsterSeeds(seed: int) void
+ setMonsterSlowDown(factor: int) : void
+ getMonsters : ArrayList<Monster>
+ getMonsterLocations(): Map<Monster,Location>

<<abstract>> Monster
- visitedList : ArrayList<Location>
- listLength : int
- randomiser : Random
- stopMoving : boolean
- isFurious : boolean
-seed:int

+ setStopMoving(seconds: int) : void
+ act() : void
+ walkApproach() : void
+ monsterWalk(oldDirection: double, next: Location) : void
+ randomWalk(oldDirection: double, next: Location) : void
- addVisitedList(location: Location) : void
+ isVisited(location: Location) : boolean
+ hasJustComeFromPair(portal : Portal) : boolean
+ teleport(portal: Portal) : void
+ steppedOnPortal() : portal
+ canMove(location: Location) : boolean
+ getType() : MonsterType

<<enumeration>> MonsterType
- Troll
- TX5

+ getImageName() : String

Troll

+ monsterWalk(oldDirection: double, next: Location) : void

TX5

+ monsterWalk(oldDirection: double, next: Location) : void

autoplayer

<<interface>> AutoPlayerStrategy
+ playAutoPlayer(pacActor: PacActor): void

FindClosestStrategy

+ playAutoPlayer(pacActor : PacActor) : void

PortalStrategy

+ playAutoPlayer(pacActor : PacActor) : void